A tradeoff explanation of quantificational conservativity

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A quantifier Q is conservative iff $Q(L, R) \leftrightarrow Q(L, L \cap R)$, i.e., its truth depends solely on the content of its first argument. Despite being an extremely strong constraint on quantifier meaning, it appears to be a universal feature in the meaning of lexicalized determiners (Barwise & Cooper, 1981). Conservativity is arguably the most well-known semantic universals.

Several explanations for conservativity have been developed. Some previous accounts argued that universals of quantification are due to conceptual simplicity (Van De Pol, Lodder, Van Maanen, Steinert-Threlkeld, & Szymanik, 2023) or ease of learning (Steinert-Threlkeld & Szymanik, 2020). This general approach struggles to explain why nonconservative-like meanings are attested and can be learned in the verbal, nominal, or adjectival domain (e.g., 'equinumerous'). Other accounts propose explanations based on manually stipulating restrictions on lexical meanings (Keenan & Stavi, 1986), or syntactic movement (Romoli, 2015), making nonconservative quantifiers inexpressible in practice. Knowlton, Pietroski, Williams, Halberda, and Lidz (2020) argues that these accounts are in tension with psycholinguistic evidence, showing that participants group the elements in the first quantifier argument but not in the second during verification tasks. Knowlton, Pietroski, Williams, Halberda, and Lidz (2023) supports an alternative account based of *restricted quantification*, which proposes that determiners do not express a relation between sets as proposed in generalized quantification theory, but rather a monadic quantifier in a universe restricted by the first argument. While restricted quantification accounts for the data, it raises the underlying evolutionary questions: why did restricted quantification evolve and not the generalized quantification of standard accounts?

We propose a novel account of the evolution of conservativity that addresses this question. In line with much previous work (Uegaki, 2024; Regier, Kemp, & Kay, 2015; Kirby, Tamariz, Cornish, & Smith, 2015), we propose that conservativity emerges as an adaptation to pressures for conceptual simplicity and communicative accuracy. We model both dimensions of optimization in a novel way. In line with previous work, the space of possible languages is specified via

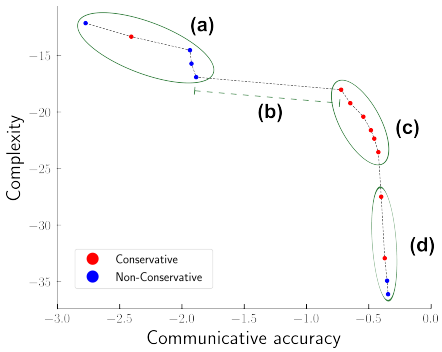

Figure 1. Pareto frontier of languages with pragmatic communication.

a formal grammar (Piantadosi & Jacobs, 2016; Regier et al., 2015; Denić & Szymanik, 2024). However, in contrast to previous models we evolve not only lexical meanings, but also a composition function that composes them together, which can contain abstractions that get reused across quantifiers. We call the associated measure of complexity *global complexity*. We show that conservativity evolves only under a joint pressure from simplicity and communicative accuracy.

The calculation of communicative accuracy includes three innovations. First, signals are not holistic, but rather compositionally structured. Second, agents do not search a pre-specified set of signals, but rather explore the generations of the language's grammar (with a depth bound). Third, the communicative task is not referential, but rather descriptive; rather than selecting one of a set of scenes, the listener's task is to reconstruct an unknown component of the world.

Figure 1 sketches the main result of the paper. Languages on the Pareto front can be roughly divided in three groups. Group (a) includes languages that are very simple but also perform poorly for communication. The introduction of conservativity allows a large jump in communicative accuracy (b) with a small increase in complexity, resulting in Group (c) containing languages that are both simple and perform well in communication. Finally, there is a long tail (d) of much more complex languages that offer only modest increases in communicative accuracy. We show that both pragmatic communication and global complexity are essential for the optimality of the conservativity abstraction.

This work makes two main contributions. First, it provides a novel account of the universal of conservativity. Second, it studies evolutionary tradeoffs in a model of compositional meaning that strongly resembles the picture of modern formal semantics (Heim & Kratzer, 1998). In this way, we hope to further strengthen the connection between evolutionary models of linguistic universals and formal semantics.

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